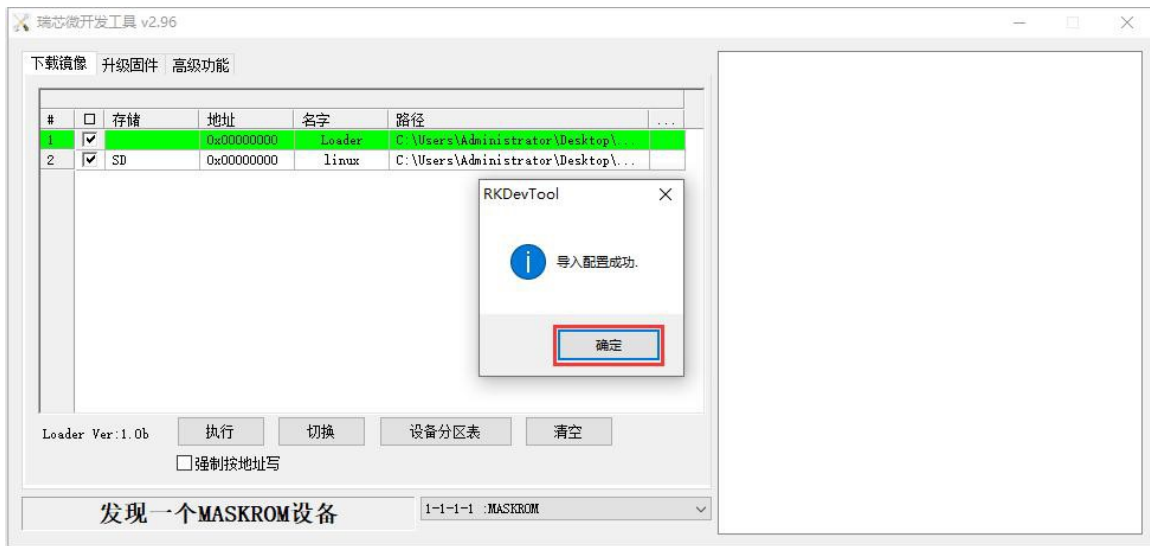
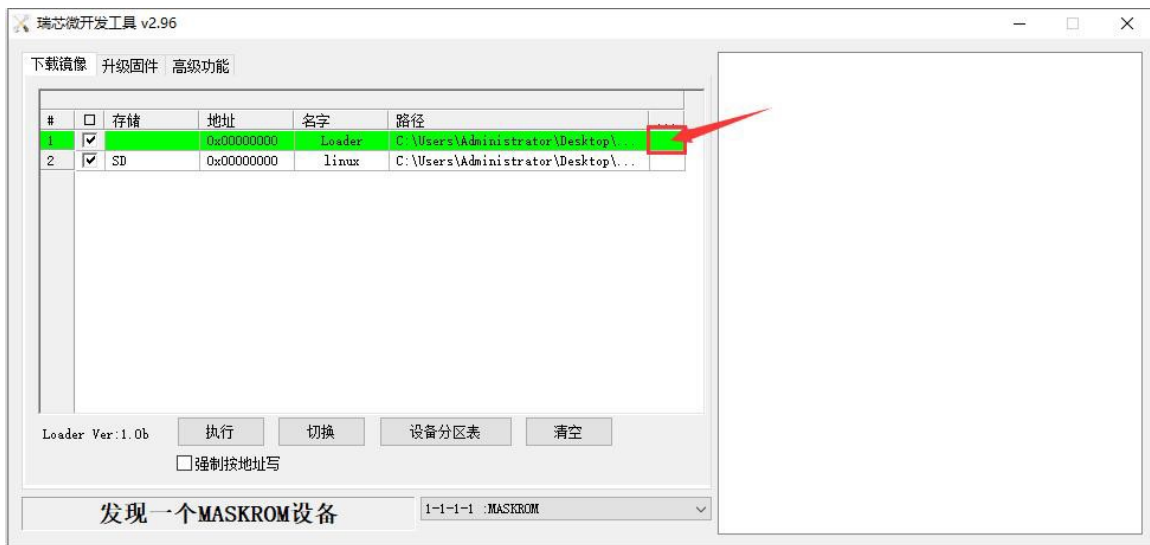




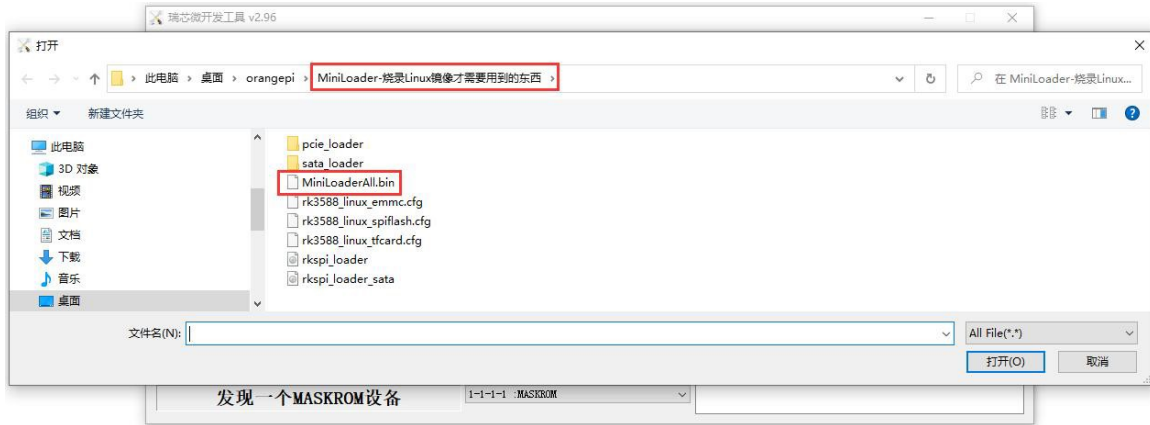
k. Then click **OK**



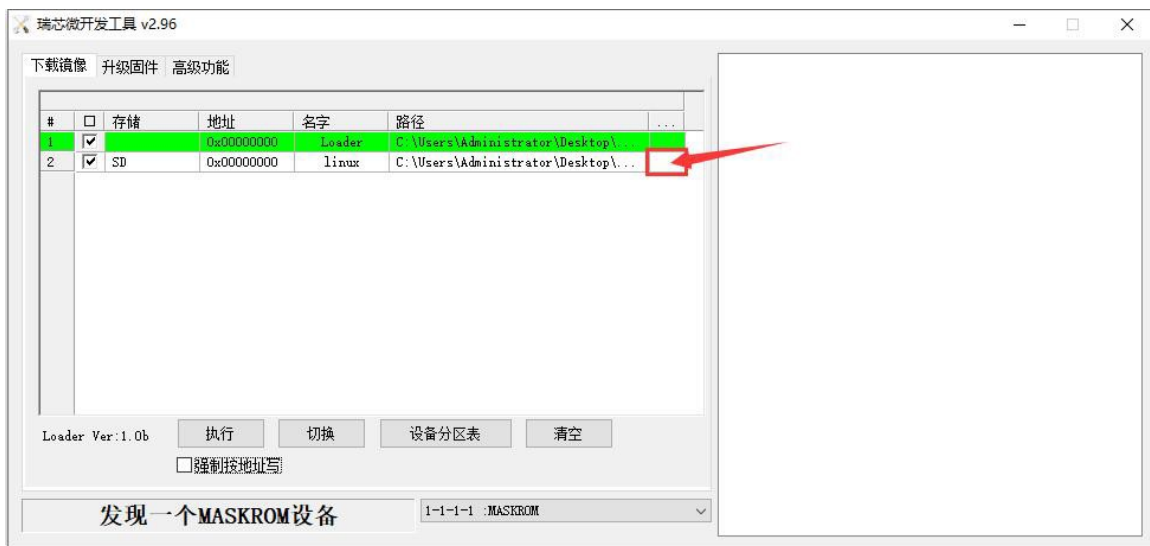
l. Then click the position shown in the figure below



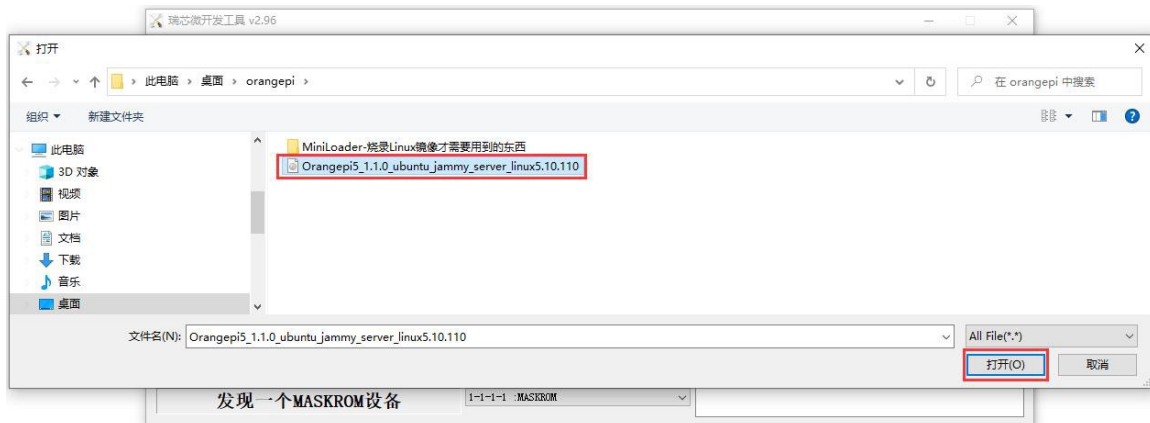
m. Select **MiniLoaderAll.bin** in the **MiniLoader** folder downloaded earlier, and then click to **open**



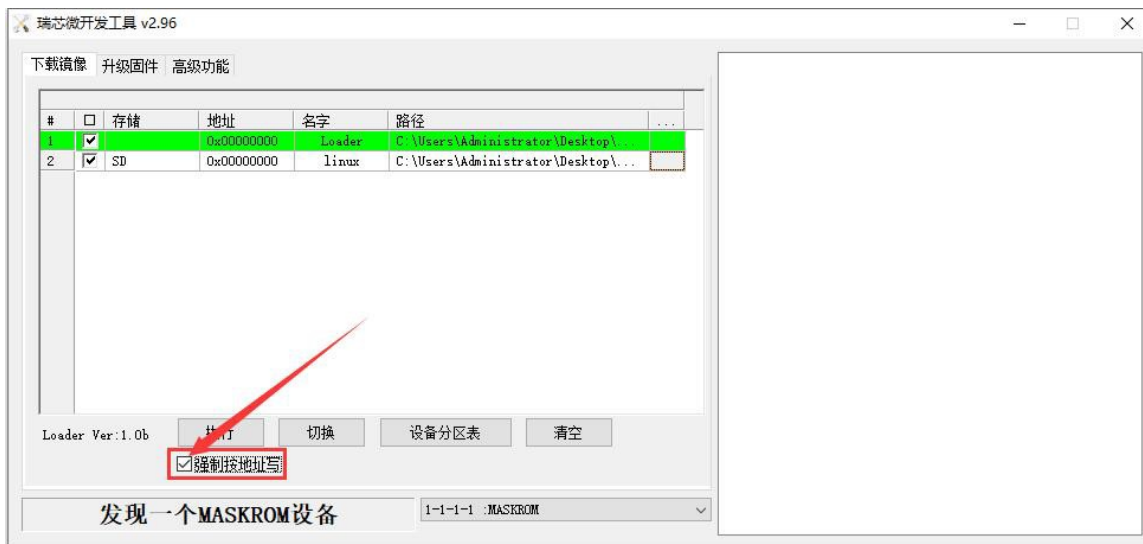
n. Then click the position shown in the figure below



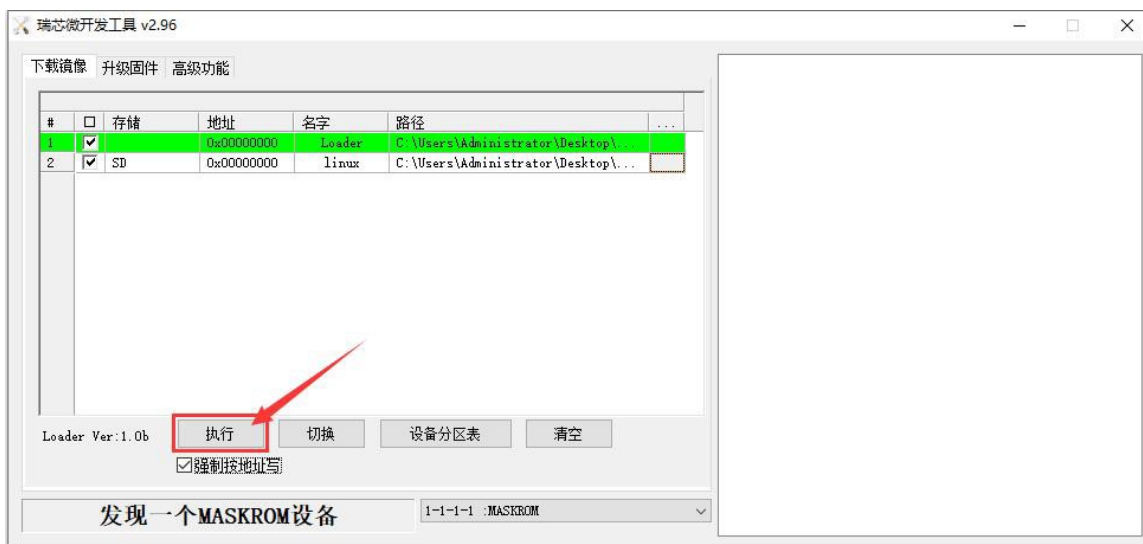
o. Then select the path of the linux image you want to burn, and then click **Open**



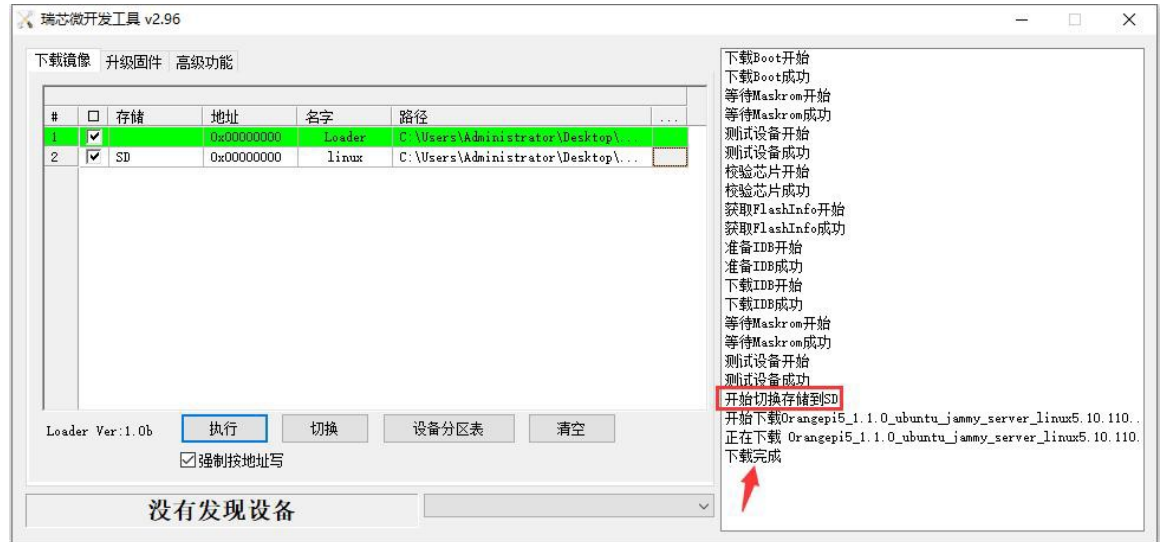
p. Then please check the option to **force write by address**



- q. Click the execute button again to start burning the Linux image to the TF card of the development board



- r. The display log after burning the Linux image is shown in the figure below 显示 log



- s. After burning the linux image to the TF card, the linux system will start automatically.

2. 3. 3. How to use Win32Diskimager to burn Linux image

1) First prepare a TF card with a capacity of 16GB or more. The transmission speed of the TF card must be **class 10** or above. It is recommended to use a TF card of SanDisk and other brands

2) Then use the card reader to insert the TF card into the computer

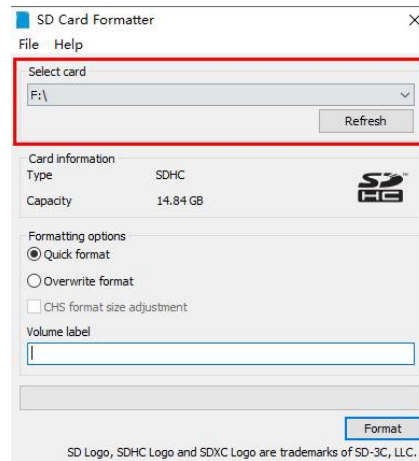
3) Then format the TF

- a. **SD Card Formatter** can be used to format the TF card. The download address is:

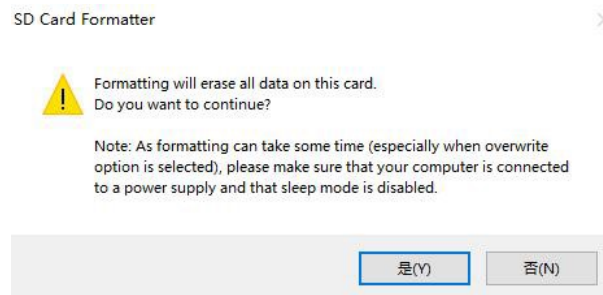
https://www.sdcard.org/downloads/formatter/eula_windows/SDCardFormatterv5_WinEN.zip

b. After downloading, unzip and install directly, and then open the software

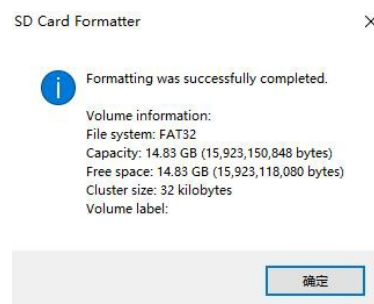
c. If only a TF card is inserted into the computer, the drive letter of the TF card will be displayed in the "Select card" column. If multiple USB storage devices are inserted into the computer, you can select the corresponding drive letter of the TF card through the drop-down box.



- d. Then click **"Format"**, a warning box will pop up before formatting, and formatting will start after selecting **"Yes (Y) "**



- e. After formatting the TF card, the information shown in the figure below will pop up, click OK



4) Then download the Linux operating system image file compression package that you want to burn from the [Orange Pi data download page](#), and then use the decompression software to decompress it. Among the decompressed files, the file ending with **".img"** is the image file of the operating system , the size is generally above 2GB

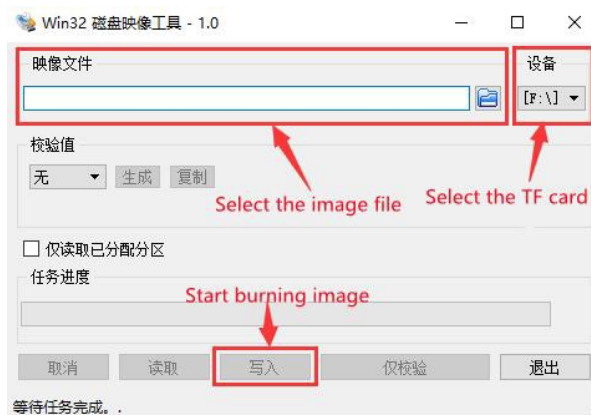
5) Use **Win32Diskimager** to burn the Linux image to the TF Card



- a. The download page of Win32Diskimager is

<http://sourceforge.net/projects/win32diskimager/files/Archive/>

- b. After downloading, install it directly. The interface of **Win32Diskimager** is as follows
- First select the path of the image
 - Then confirm that the drive letter of the TF card is consistent with that displayed in the "**Device**" column
 - Finally click "**Write**" to start burn



- c. After the image writing is completed, click the "**Exit**" button to exit, and then you can pull out the TF card and insert it into the development board to start

2. 4. How to burn Linux image to TF card based on Ubuntu

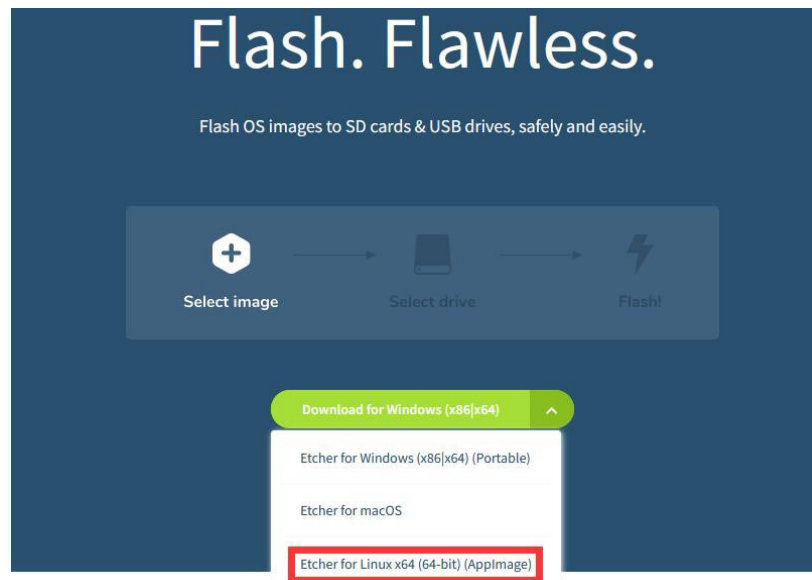
Note that the Linux image mentioned here specifically refers to the image of Linux distributions such as Debian or Ubuntu downloaded from the Orange Pi data download page, and the Ubuntu PC refers to the personal computer with the Ubuntu system installed.

- 1) First prepare a TF card with a capacity of 16GB or more. The transmission speed of the TF card must be **class 10** or above. It is recommended to use a TF card of SanDisk and other brand
- 2) Then use the card reader to insert the TF card into the computer
- 3) Download the balenaEtcher software, the download address is:

<https://www.balena.io/etcher/>



4) After entering the balenaEtcher download page, please select the Linux version of the software from the drop-down box to download



5) Download the Linux operating system image file compression package that you want to burn from the [Orange Pi data download page](#), and then use the decompression software to decompress it. Among the decompressed files, the file ending with ".img" is the image file of the operating system. The size is generally more than 22GB

The decompression command for the compressed package ending in 7z is as follow

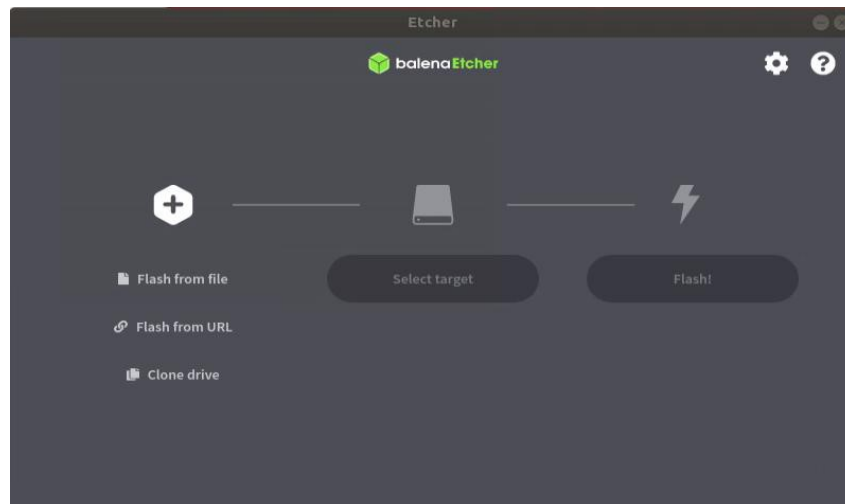
```
test@test:~$ 7z x Orangepi5_1.0.0_debian_bullseye_desktop_xfce_linux5.10.110.7z
test@test:~$ ls Orangepi5_1.0.0_debian_bullseye_desktop_xfce_linux5.10.110.*
Orangepi5_1.0.0_debian_bullseye_desktop_xfce_linux5.10.110.7z
Orangepi5_1.0.0_debian_bullseye_desktop_xfce_linux5.10.110.sha    #checksum file
Orangepi5_1.0.0_debian_bullseye_desktop_xfce_linux5.10.110.img    #Image file
```

6) After decompressing the image, you can first use the `sha256sum -c *.sha` command to calculate whether the checksum is correct. If the prompt is **successful**, it means that the downloaded image is correct, and you can safely burn it to the TF card. If it prompts that the **checksum does not match**, it means There is a problem with the downloaded image, please try

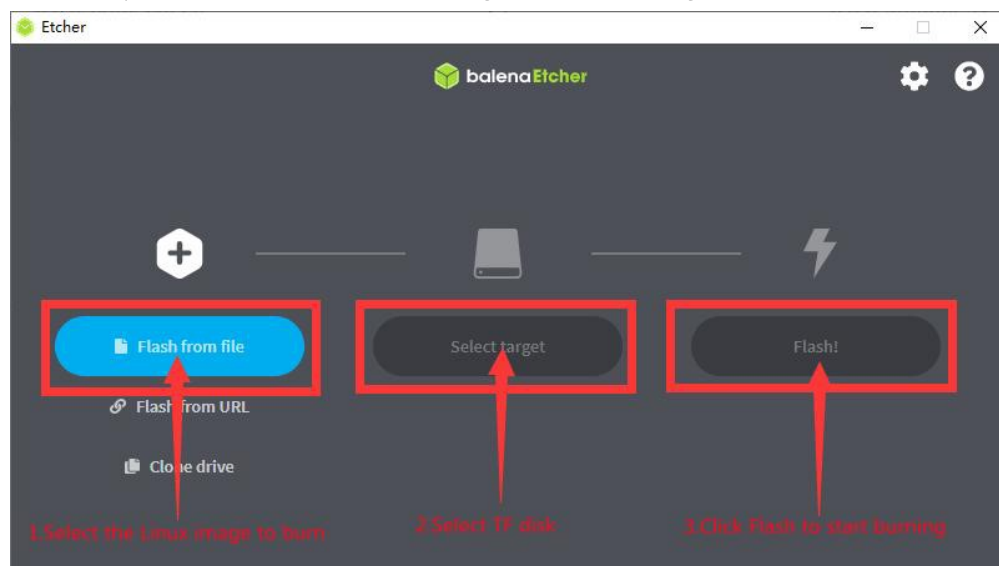
```
test@test:~$ sha256sum -c *.sha
Orangepi5_1.0.0_debian_bullseye_desktop_xfce_linux5.10.110.img: OK
```



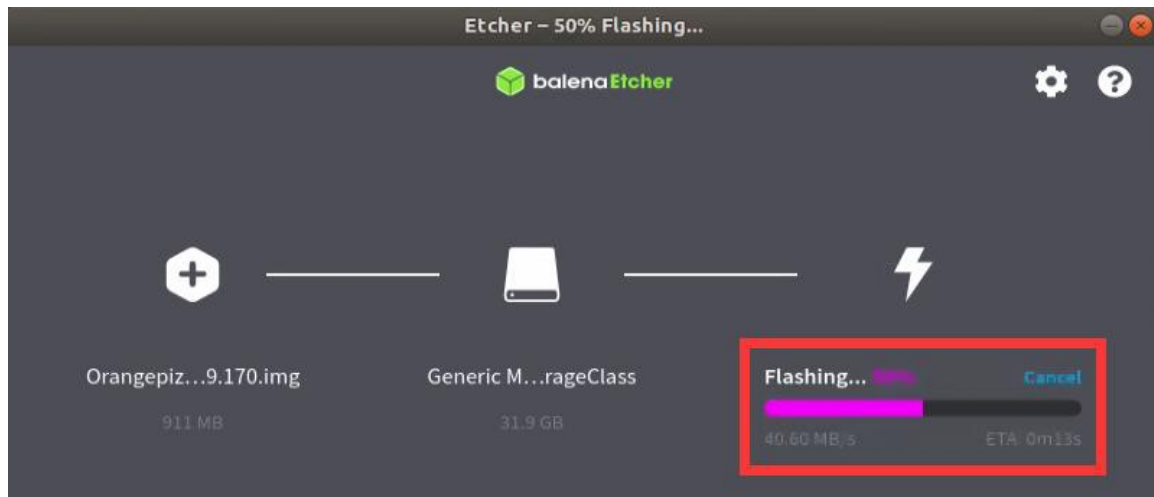
7) Then double-click **balenaEtcher-1.5.109-x64.AppImage** on the graphical interface of Ubuntu PC to open balenaEtcher (**no installation required**), and the interface after balenaEtcher is opened is shown in the figure below



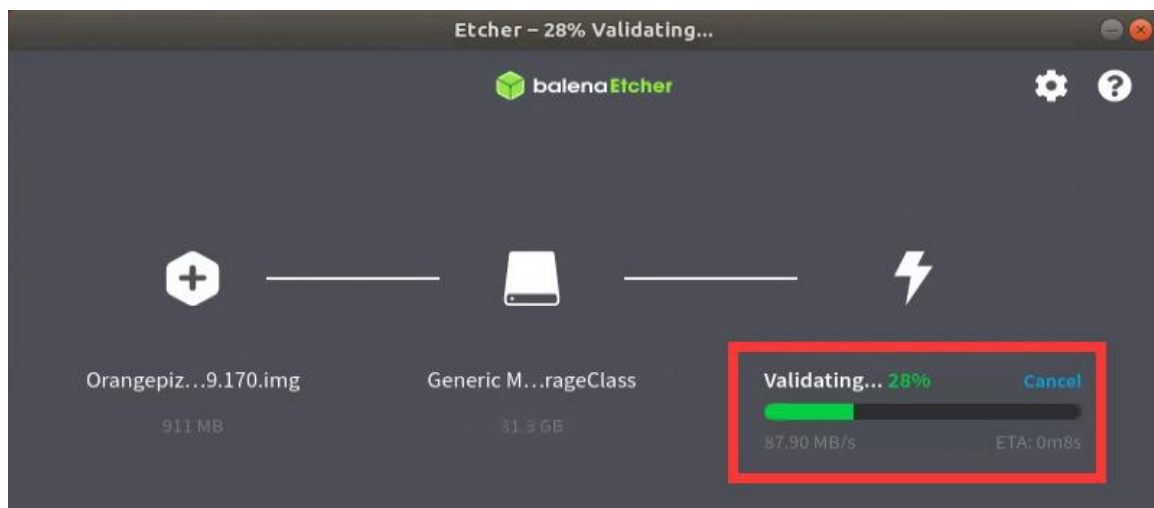
- 8) The specific steps to use balenaEtcher to burn the Linux image are as follows
- First select the path of the Linux image file to be burned
 - Then select the drive letter of the TF Card
 - Finally, click Flash to start burning the Linux image to the TF Card



9) The interface displayed during the process of burning the Linux image by balenaEtcher is shown in the figure below, and the progress bar displays purple, indicating that the Linux image is being burned into the TF Card



11) After burning the Linux image, balenaEtcher will also verify the image burned into the TF card by default to ensure that there is no problem in the burning process. As shown in the figure below, a green progress bar indicates that the image has been burnt, and balenaEtcher is verifying the burnt image.



12) The display interface of Balenaetcher after the successful record is completed. If the green indicator icon is displayed in the figure below, the image burning is successful, then you can exit Balenaetcher, then unplug the TF card into the TF card slot in the development board and use it.



2. 5. How to write Linux image to SPI Flash+NVMe SSD

Note that the Linux image mentioned here specifically refers to the images of Linux distributions such as Debian or Ubuntu downloaded from the Orange Pi data download page.

Note that all the following operations are performed on a Windows computer.

2. 5. 1. How to use RKDevTool to burn

1) First, you need to prepare an NVMe SSD. The PCIe supported by the M.2 slot of the development board is PCIe2.0x1, and the theoretical maximum speed is 500MB/s. PCIe3.0 and PCIe4.0 NVMe SSDs are also available, but the highest speed is only PCIe2.0.

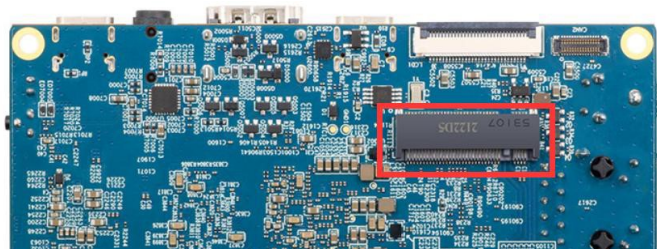
a. The M.2 2230 SSD is as follows



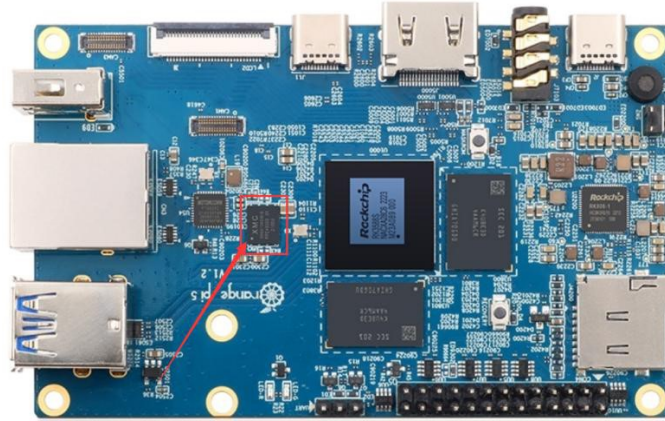
b. The M.2 2242 SSD is as follows



2) Then insert the NVMe SSD into the M.2 PCIe interface of the development board and fix it



3) The position of the SPI Flash on the development board is shown in the figure below, no other settings are required before starting the programming

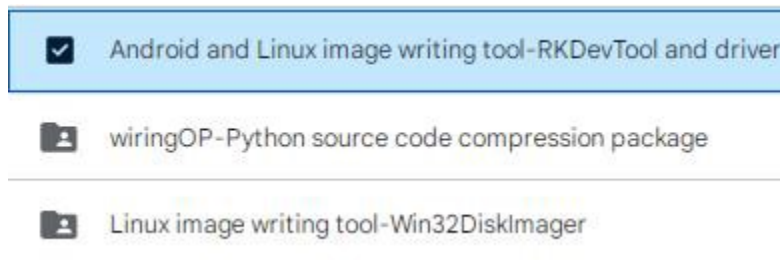


4) Then you need to prepare a data cable with good quality Type-C interface

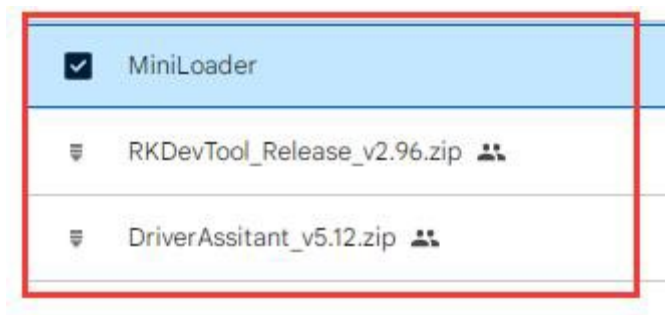


5) Then download Rockchip **DriverAssitant_v5.12.zip** and **MiniLoader** and the burning tool **RKDevTool_Release_v2.96.zip** from the [Orange Pi data download page](#), please make sure that the version of the downloaded **RKDevTool** tool is **v2.96**

- a. On the Orange Pi data download page, first select the official tool, and then enter the following folder



- b. Then download all the files below



Note that the folder of MiniLoader - the things needed to burn the Linux image is hereinafter referred to as the MiniLoader folder.

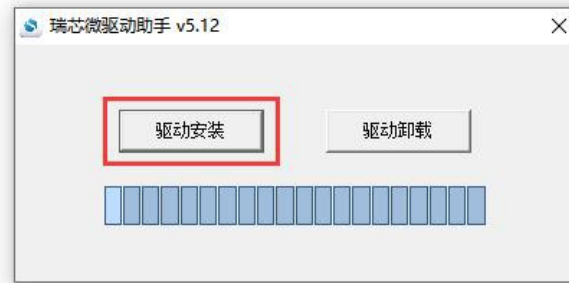
5) Then download the Linux operating system image file compression package that you want to burn from the [Orange Pi data download page](#), and then use the decompression software to decompress it. Among the decompressed files, the file ending with ".img" is the image file of the operating system , the size is generally above 2GB

6) Then use decompression software to decompress **DriverAssitant_v5.12.zip**, and then find the **DriverInstall.exe** executable file in the decompressed folder and open it

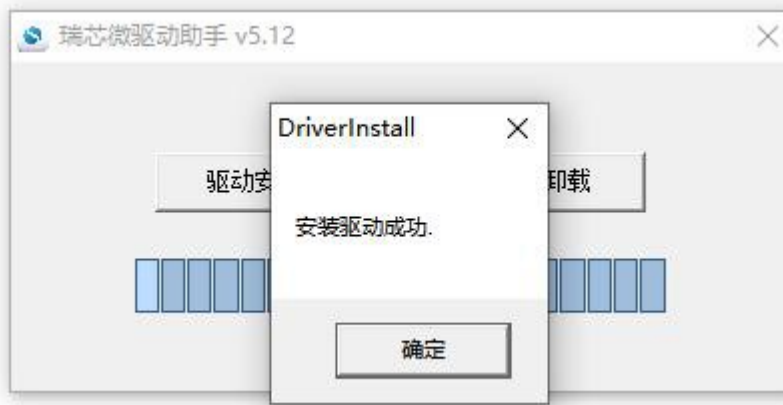
名称	修改日期	类型	大小
ADBDriver	2022/12/1 15:07	文件夹	
bin	2022/12/1 15:07	文件夹	
Driver	2022/12/1 15:07	文件夹	
config	2014/6/3 15:38	配置设置	1 KB
DriverInstall	2022/2/28 14:11	应用程序	491 KB
Readme	2018/1/31 17:44	文本文档	1 KB
revision	2022/2/28 14:14	文本文档	1 KB

7) After opening **DriverInstall.exe**, the steps to install the Rockchip driver are as follows

- a. Click the "**Driver Installation**" button



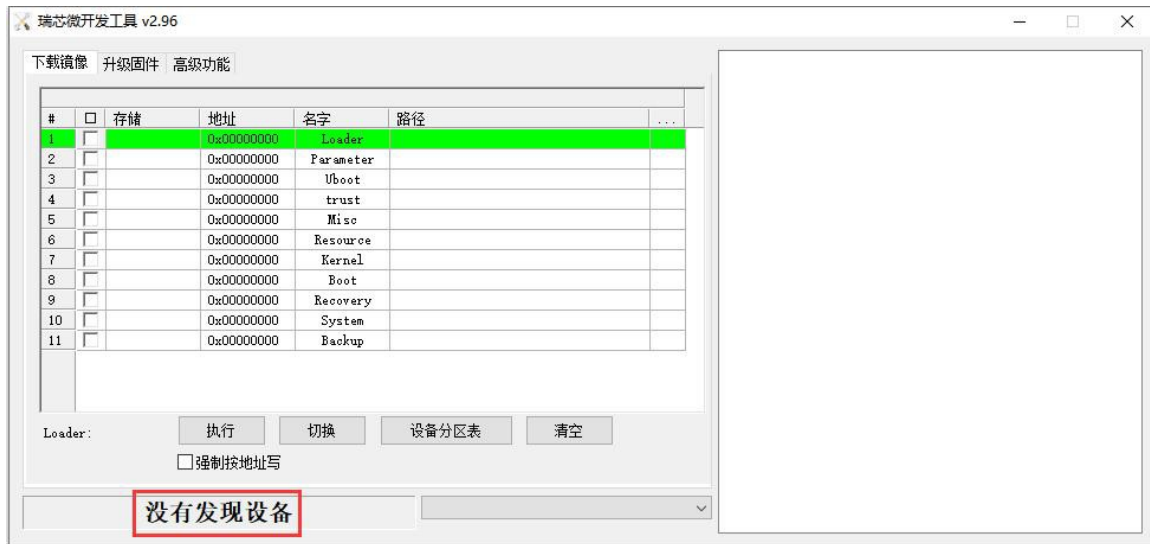
- b. After waiting for a while, a pop-up window will prompt "**driver installed successfully**", and then click the "**OK**" button.



- 8) Then decompress **RKDevTool_Release_v2.96.zip**, this software does not need to be installed, just find **RKDevTool** in the decompressed folder and open it

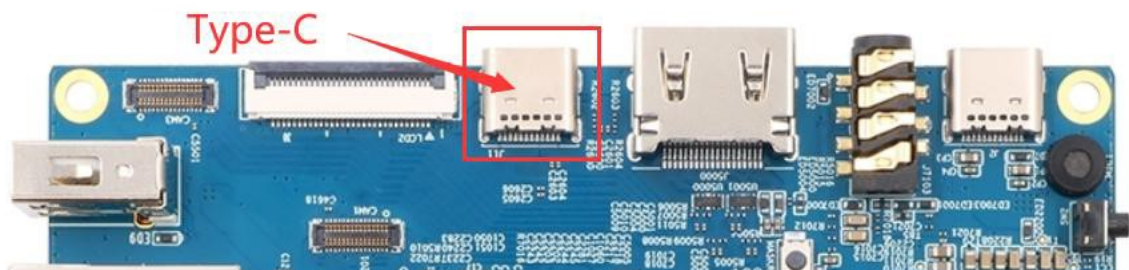
名称	修改日期	类型	大小
bin	2022/12/1 15:07	文件夹	
Language	2022/12/1 15:07	文件夹	
config.cfg	2022/3/23 9:11	CFG 文件	7 KB
config	2021/11/30 11:04	配置设置	2 KB
revision	2022/5/27 9:09	文本文档	3 KB
RKDevTool	2022/5/27 9:06	应用程序	1,212 KB
开发工具使用文档_v1.0	2021/8/27 10:28	Foxit PDF Reade...	450 KB

- 9) After opening the **RKDevTool** burning tool, because the computer has not been connected to the development board through the Type-C cable at this time, the lower left corner will prompt "**No device found**"

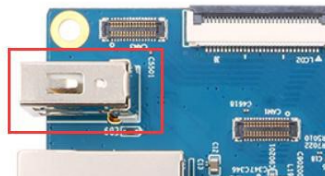


10) Then start burning the Linux image to the SSD

- a. First, connect the development board to the Windows computer through the Type-C data cable. The position of the Type-C interface on the development board is shown in the figure below.



- b. Make sure that the development board is not connected to the power supply and inserted into the TF card.
- c. Also make sure that the white USB2.0 interface in the position shown below is not plugged into a USB device USB2.0.



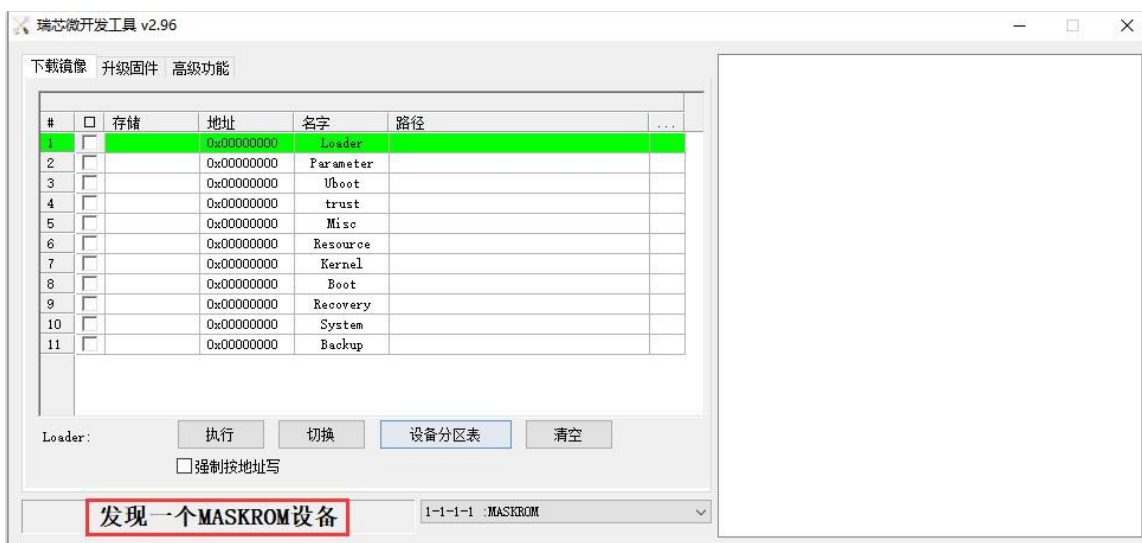
- d. Then press and hold the MaskROM button on the development board. The position of the MaskROM button on the development board is shown in the figure below:



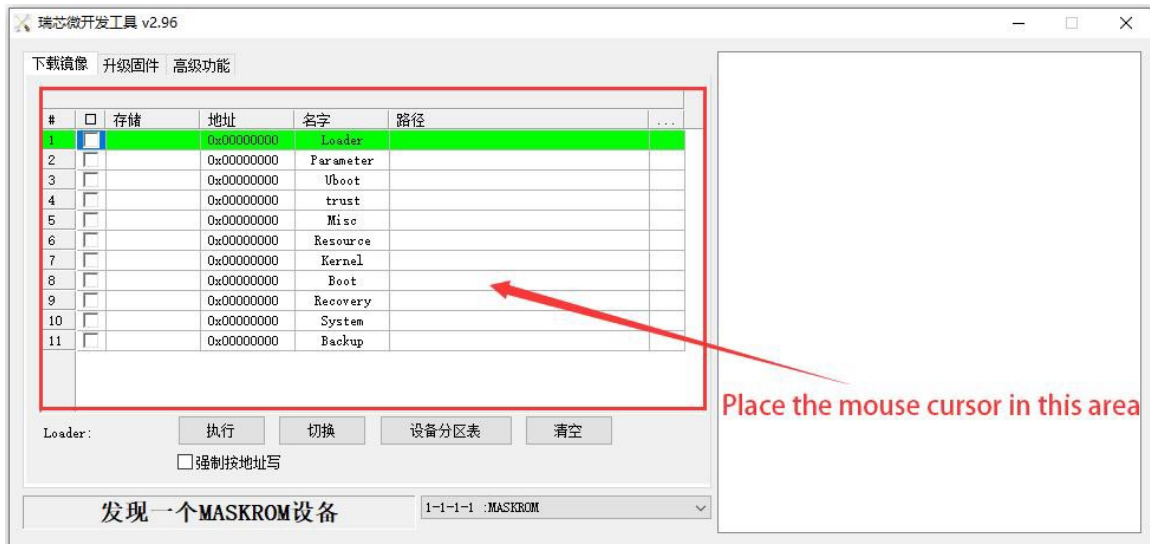
- e. Then connect the power supply of the Type-C interface to the development board, and power on, and then release the MaskROM button



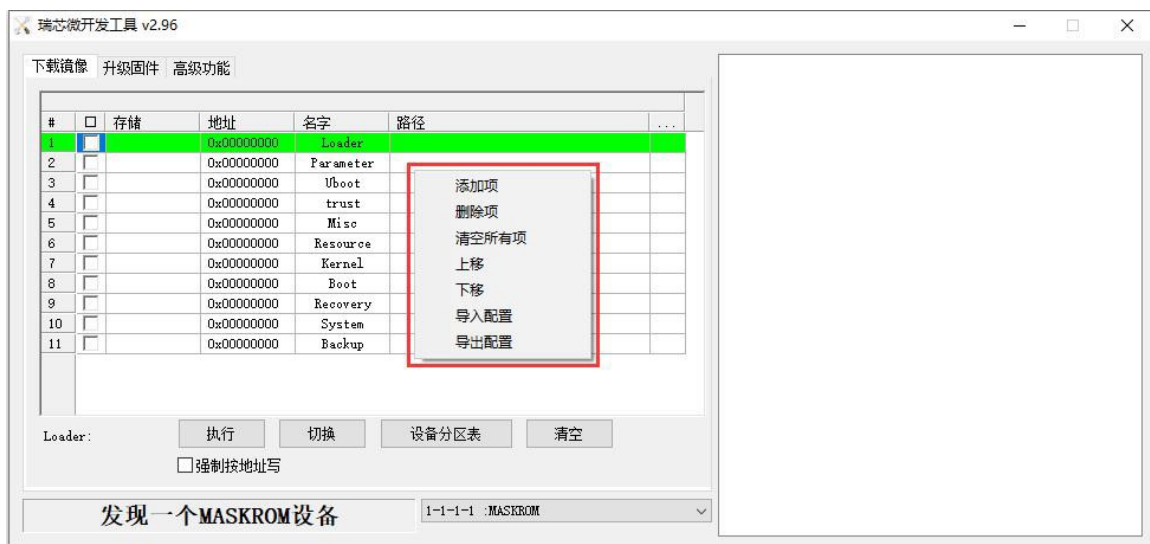
- f. If the previous steps are successful, the development board will enter the **MASKROM** mode at this time, and the interface of the burning tool will prompt "found a MASKROM device"



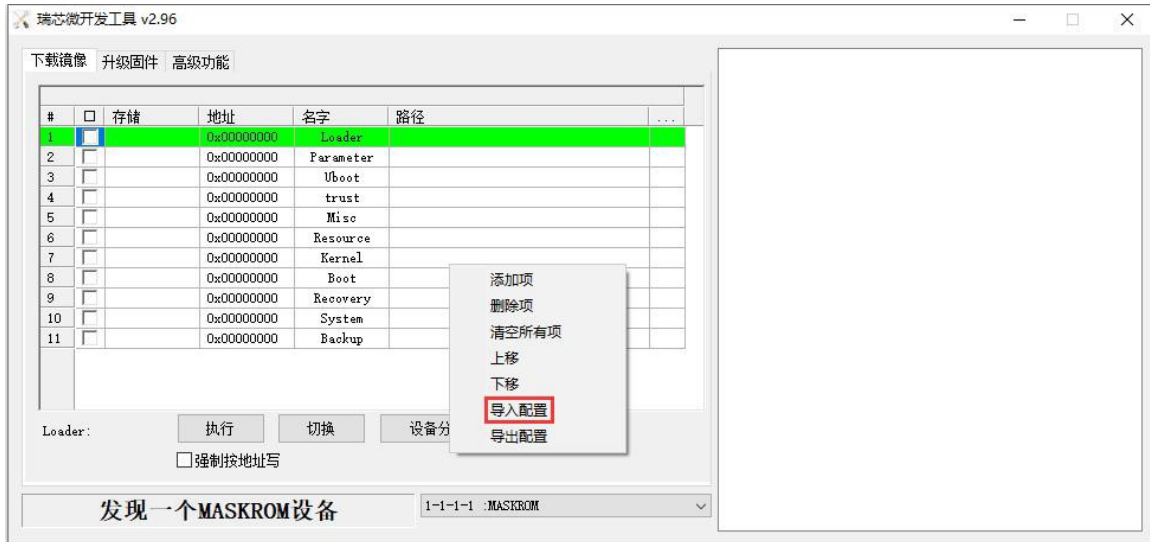
- g. Then place the mouse cursor in the area below



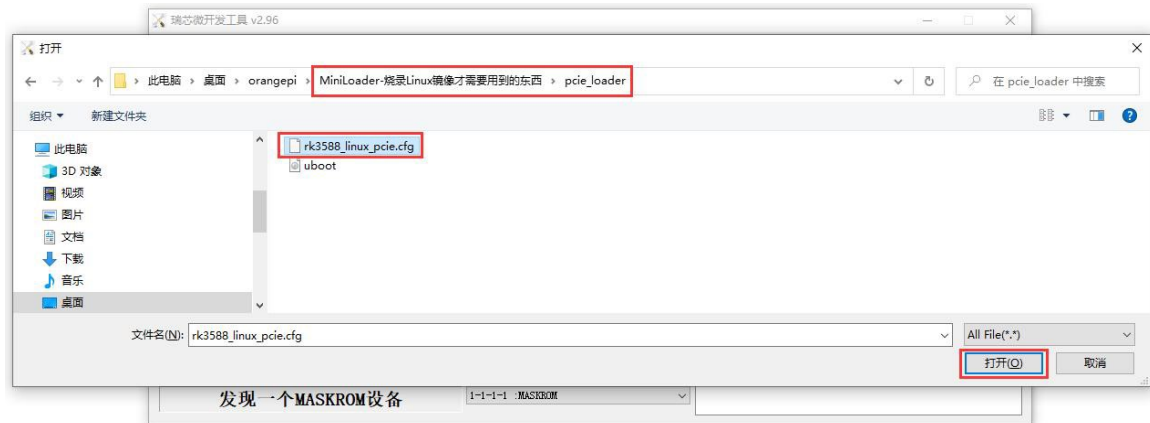
- h. Then click the right mouse button and the selection interface shown in the figure below will pop up



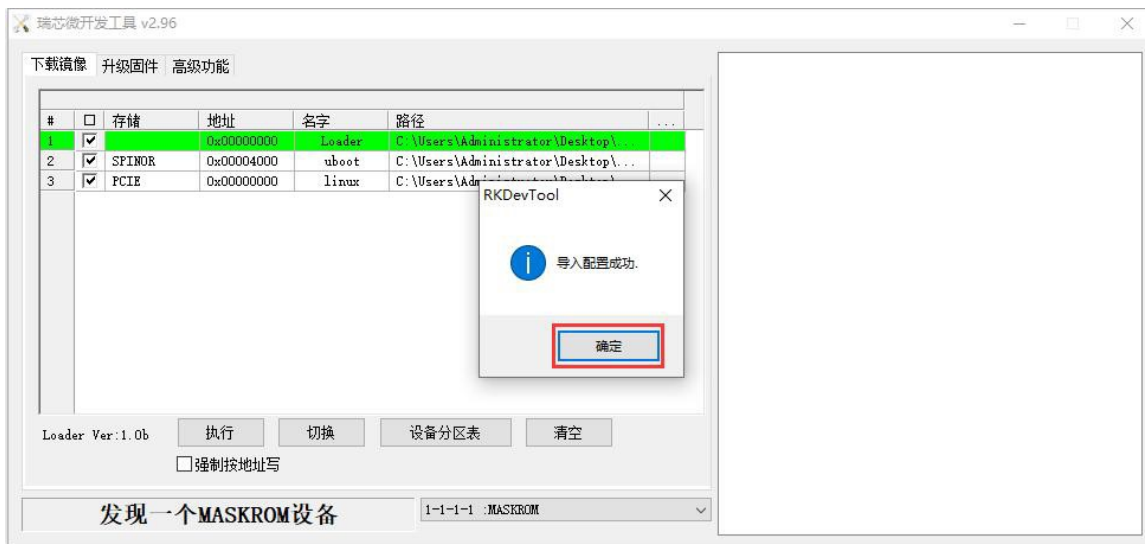
- i. Then select the **import configuration** option



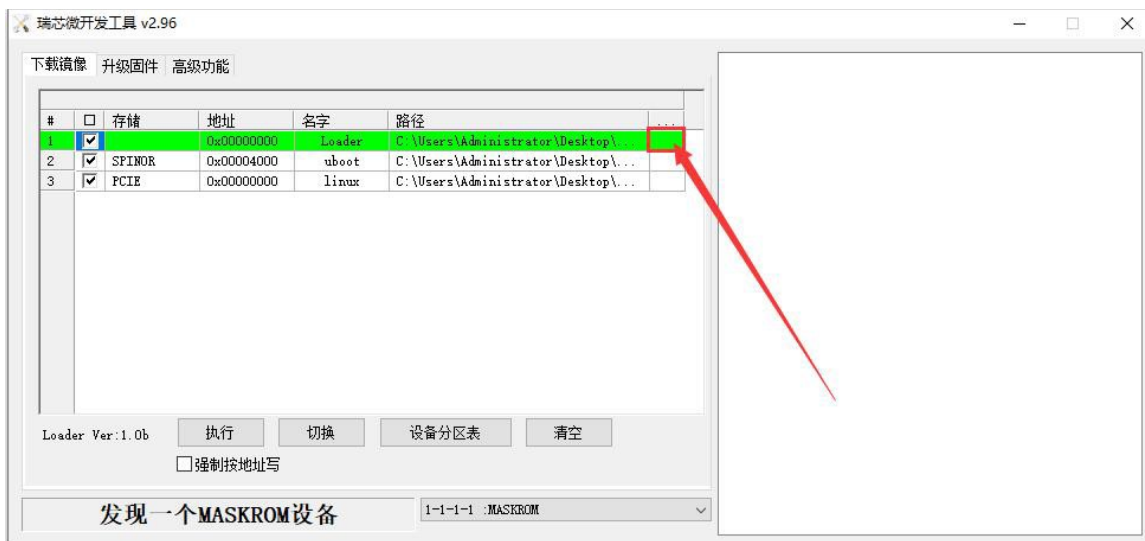
- j. Then enter the **MiniLoader** folder downloaded earlier, and then enter the pcie_loader folder, then select the **rk3588_linux_pcie.cfg** configuration file, and click **Open**.



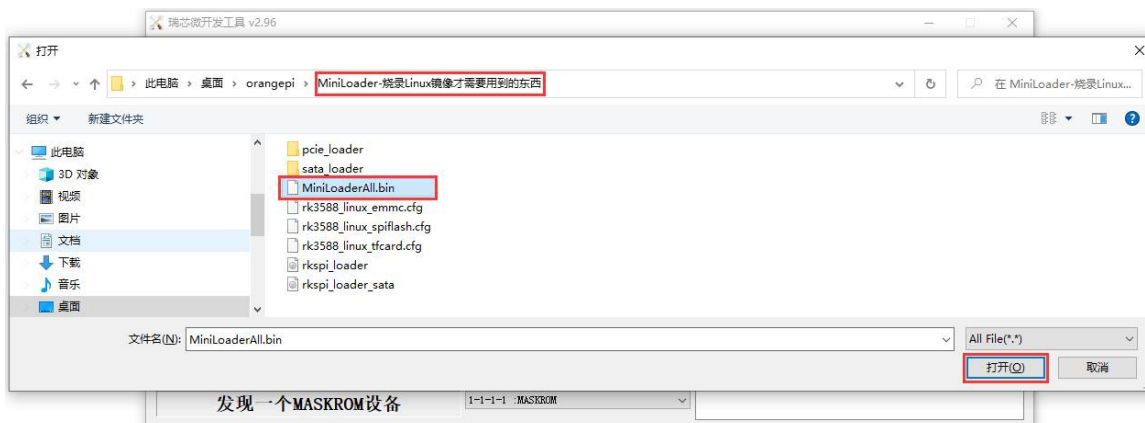
- k. Then click **OK**



1. Then click the position shown in the figure below

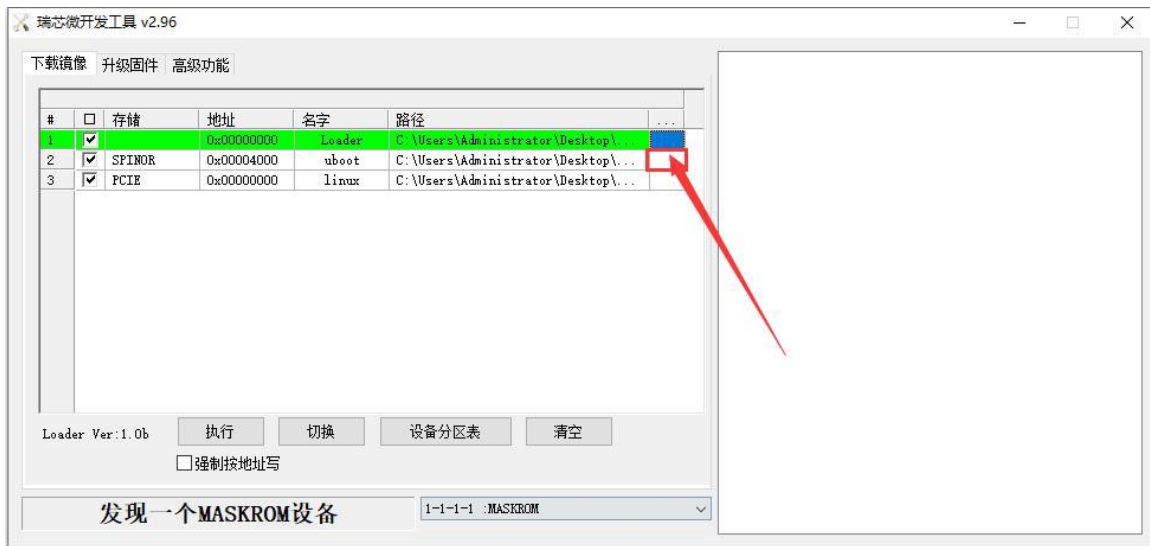


m. Select **MiniLoaderAll.bin** in the **MiniLoader** folder downloaded earlier, and then click to open

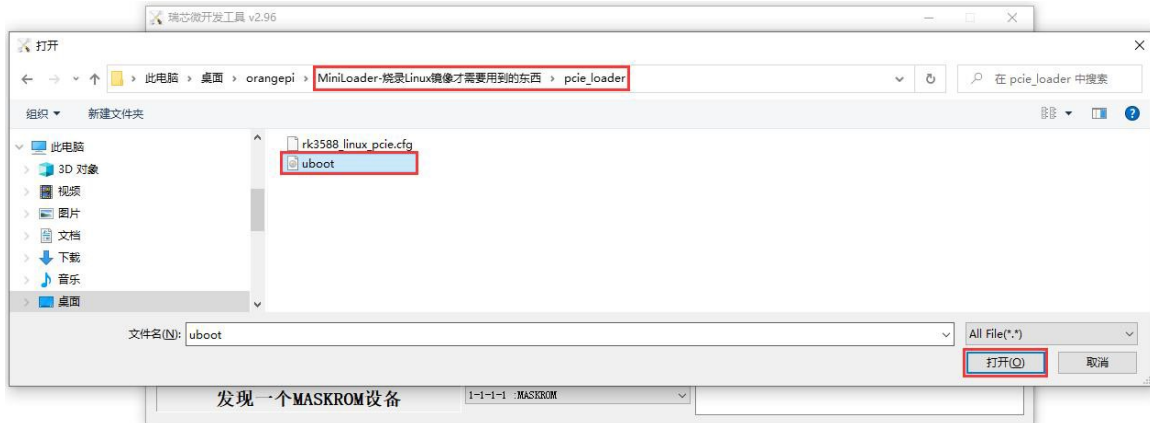




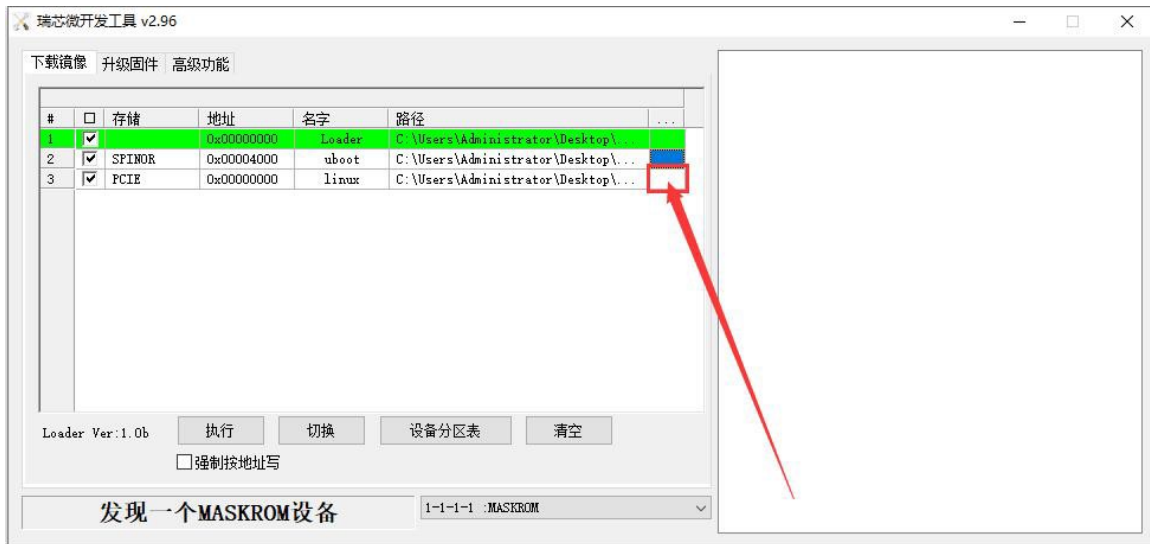
n. Then click the position shown in the figure below



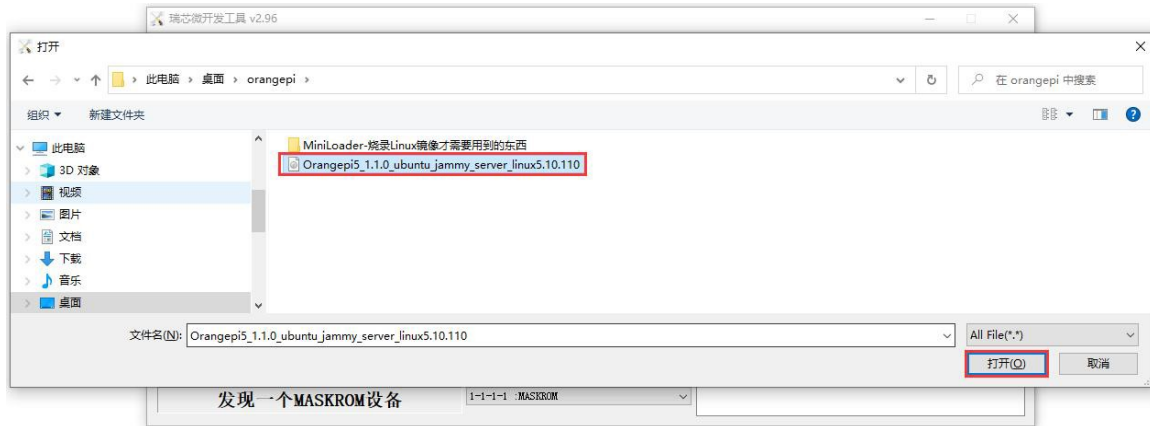
o. Then enter the **MiniLoader** folder downloaded earlier, then enter the **pcie_loader** folder, then select the **uboot** file, and click **Open**.



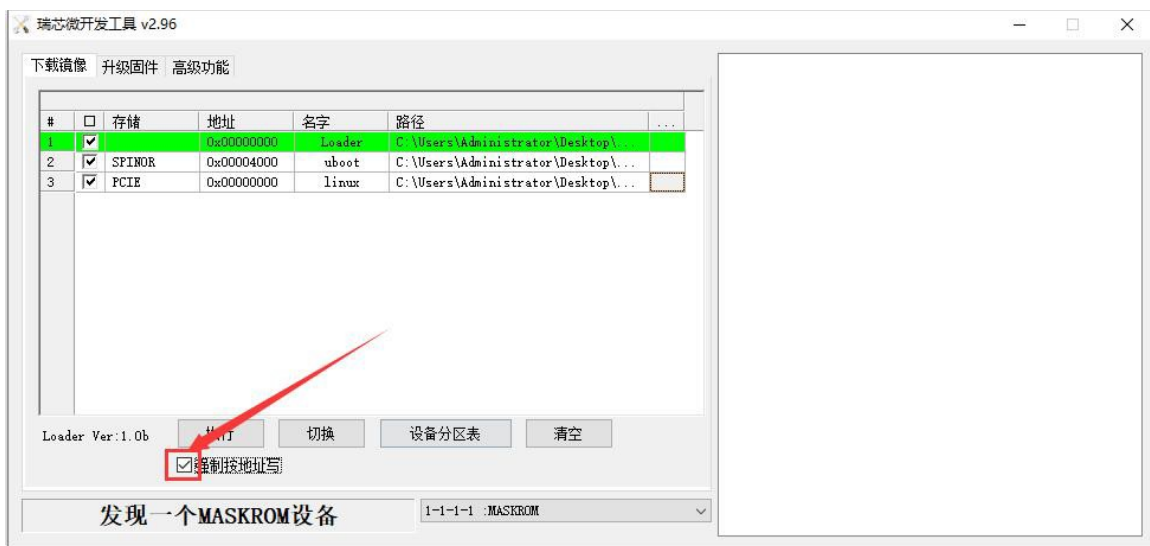
p. Then click the position shown in the figure below



q. Then select the path of the linux image you want to burn, and click **Open**

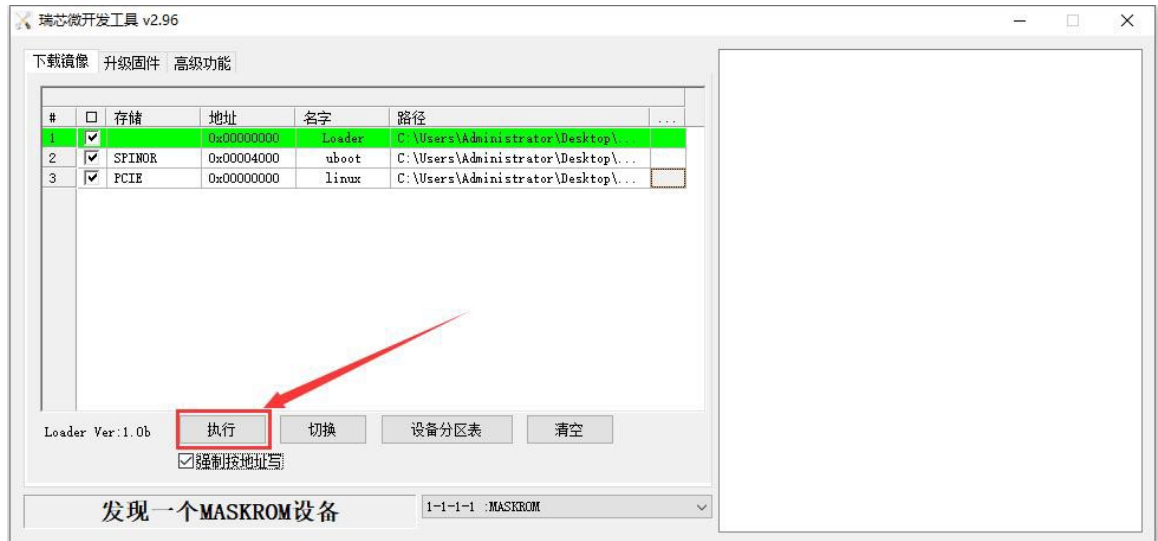


r. Then please check the option to **force writing by address**

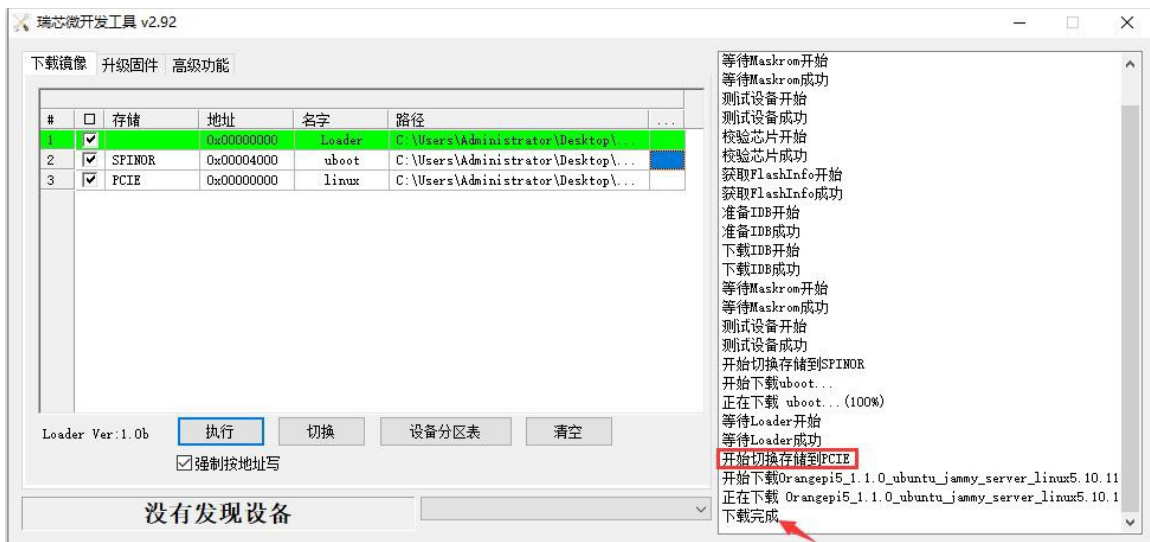




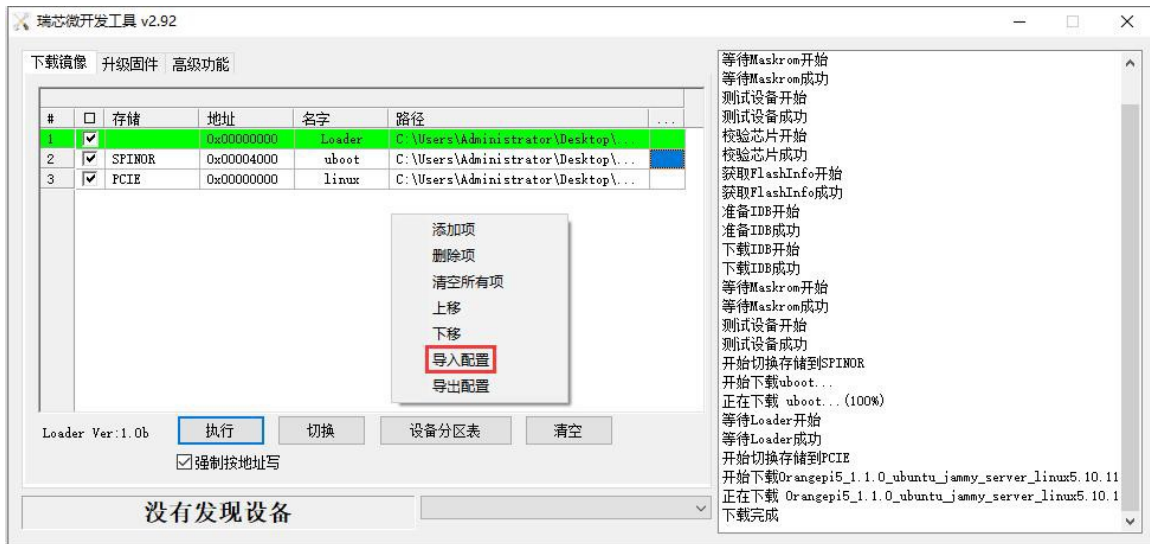
- s. Click the Execute button again to start burning the linux image to the SSD



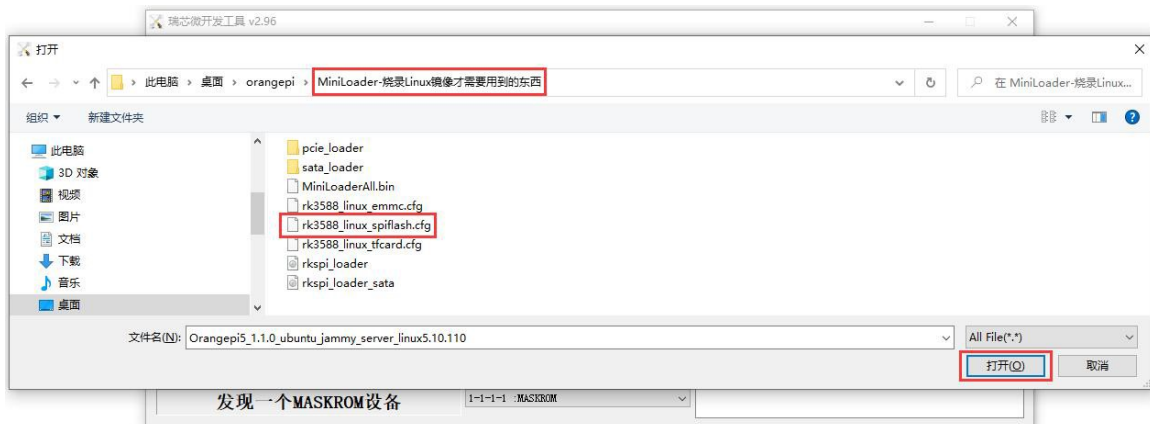
- t. The log displayed after burning the linux image is shown in the figure below



- u. Then burn the uboot image required by the linux system in the SSD to SPIFlash, and **first enter the MaskROM mode again.**
- v. Then select the **import configuration** option



- w. Then select the **rk3588_linux_spiflash.cfg** configuration file in the **MiniLoader** folder downloaded earlier, and click **Open**.



- x. Then click **OK**

